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Countability and Separation Axioms

1.1 The Countability Axioms

Definition 1.1.1 ► First-Countable Space

A space X is said to have a **countable basis at x** if there is a countable collection $\mathcal{B}$ of neighborhoods of x such that each neighborhood of x contains at least one of the elements of $\mathcal{B}$. A space that has a countable basis at each of its points is said to satisfy the **first countability axiom**, or to be **first-countable**.

Definition 1.1.2 ► Second-Countable Space

topological space X is called second-countable if it has a countable basis for its topology.

Definition 1.1.3 ► Lindelöf Space

A topological space X is called a Lindelöf space if every open cover of X has a countable subcover.

Theorem 1.1.4 ► Lindelöf Theorem

Every second-countable space is Lindelöf.

Proof. Let X be a second-countable space, $\mathcal{B} = \{B_n\}_{n=1}^{\infty}$ is a countable basis of X . Let $\mathcal{U} = \{U_{\alpha}\}_{\alpha \in A}$ be any open cover of X . Let

$$\mathcal{B}' := \{B_n \in \mathcal{B} : \exists U_{\alpha} \in \mathcal{U}, \text{ s.t. } B_n \subseteq U_{\alpha}\}$$

For each $x \in X$, $\exists U_{\alpha} \in \mathcal{U}$ such that $x \in U_{\alpha}$. Since $\mathcal{B}$ is a basis, there $\exists B_n \in \mathcal{B}$ such that $x \in B_n \subseteq U_{\alpha}$. $B_n \in \mathcal{B}'$ by definition. Thus $\mathcal{B}'$ covers X . For each $B_{n_k} \in \mathcal{B}'$, we choose an open set U_k in $\mathcal{U}$, such that $B_{n_k} \subseteq U_k$. Then $\{U_k\}_{k=1}^{\infty}$ is a subcover of $\mathcal{U}$. $\square$

1.2 The Separation Axioms

1.3 Partition of Unity

Definition 1.3.1

Let $\{U_1, \dots, U_n\}$ be a finite indexed open covering of the space X . An indexed family of continuous functions

$$\phi_i : X \longrightarrow [0, 1] \quad \text{for } i = 1, \dots, n,$$

is said to be a **partition of unity** dominated by $\{U_i\}$ if:

(1) $\text{support } \phi_i \subset U_i$ for each i .

(2) $\sum_{i=1}^n \phi_i(x) = 1$ for each x .

Theorem 1.3.2 ► Existence of finite partitions of unity

Let $\{U_1, \dots, U_n\}$ be a finite open covering of the normal space X . Then there exists a partition of unity dominated by $\{U_i\}$.

1.4 The Urysohn Metrization Theorem